

Build a BI Dashboarding Tool

Role	Product Interface Engineer
Format	Take-home assignment
Duration	1 week
Output	Working tool + architecture diagram + brief write-up

The brief

Existing BI dashboarding tools — Power BI, Looker, Holistics, QlikSense, Metabase, Tableau — all make compromises on simplicity vs. power. Some are geared towards business users and some are geared towards data/business analysts. Your job is to study these, identify UX insights, and build a focused BI dashboard tool that acts on it.

The tool should be geared towards a business user with no technical background. You can limit your analysis to Looker and any other tool geared towards a business user. You decide the scope — but everything you ship must be polished and deliberate.

A note on scope

We are not expecting a replication of Looker or Power BI. We are expecting evidence that you can think clearly about a user, make sharp design decisions, and ship something polished — even if that's a small, well-chosen subset of features done exceptionally well.

What to deliver

1. Working prototype

A published, interactive app accessible via a URL.

2. Architecture Walkthrough

Include a short architecture diagram or written explanation covering your component structure, state management approach, scalability considerations, and any other architectural components.

2. Brief Write-Up or Video

Walk us through:

- Your analysis — what you found and what you decided to act on
- Your design decisions — what you built and why
- What you would build next with another week
- Which AI tools you used and why

What we're evaluating

Criterion	What we're looking for
UX thinking	Can you identify a real user problem and solve it deliberately? Does your design show evidence of a point of view — not just features copied from existing tools?
Visual craft	Does the interface feel intentional, clean, and appropriate for a non-technical business user? Consistency, hierarchy, and usability all count.
Architecture	Does your component structure and data flow make sense? Is your architecture scalable and taken cost-considerations into account?
Scope judgment	Did you ship something real and polished in a week — not a half-finished wishlist? Sharp focus beats broad coverage.
AI fluency	Did you use AI tools to move faster and produce better output? Where, and how? This is expected.

Constraints

- Must be published and accessible via a URL
- No code or SQL required from the end user: Must be geared towards a business user, not business analyst
- Tech stack: The frontend must be built in React. Use Plotly for all chart rendering. Beyond that, stack choices are yours — your decisions around state management, component libraries, and build tooling are part of what we're evaluating. Be prepared to explain them in your write-up.

On using AI tools

Using AI tools is expected. We're not testing whether you can build without them. We're evaluating whether you can direct them well, make good judgments about what they produce, and move faster because of them.